

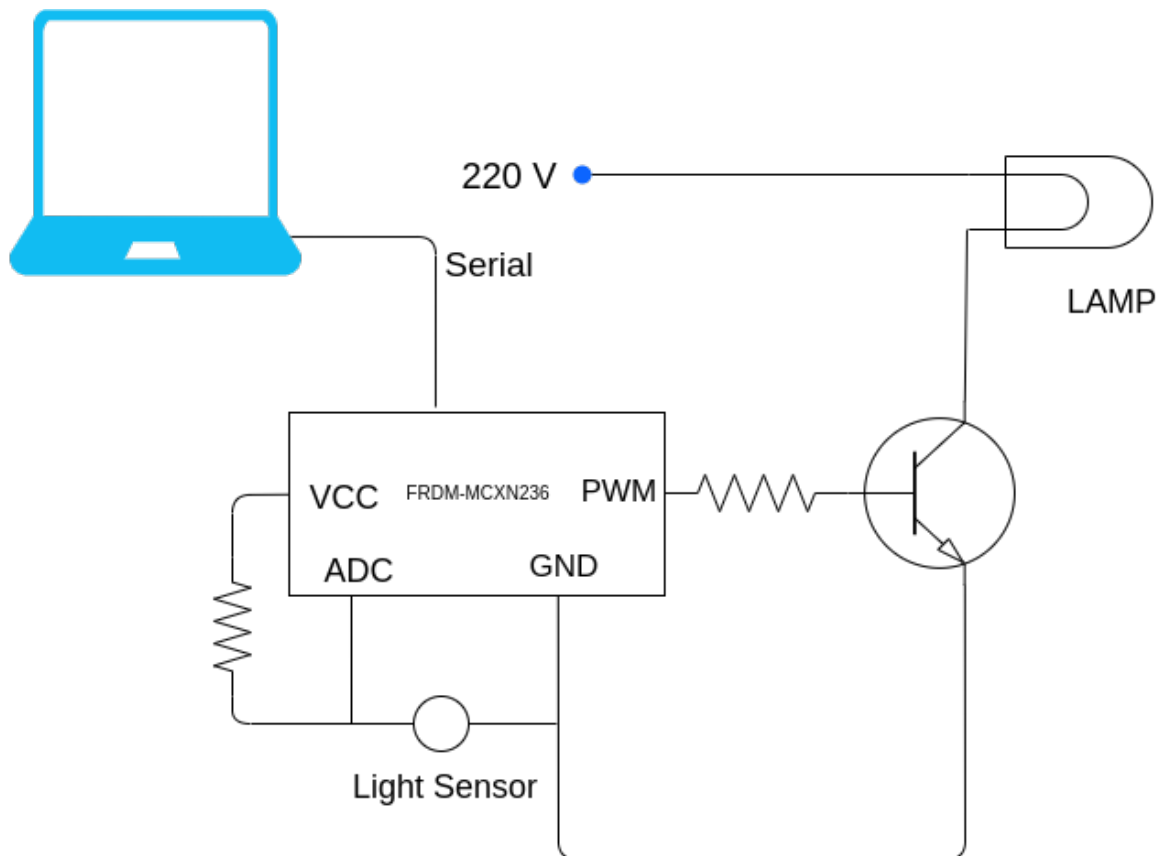
Example 1

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Lyssna

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A system to try to keep light intensity in my room at a configurable value(lux).



Environment	Footcandles	Lux
Living Room	10–20	108–215
Bedrooms	20–50	215–538
Kitchen-General	20–50	215–538
Kitchen-Preparing/Cooking	50–100	538–1,076
Dining Room	10–20	108–215
Rooms for Difficult Reading & Writing	50–100	538–1,076
Bathrooms	20–50	215–538
Hallways	5–10	54–108
Basement/Garage Workshop	20–75	215–807
Basement-Laundry	20–100	215–1,076
Basement-Exercise and Recreation	30	323

Stakeholder Requirements:

ID: REQ-01

Description: The system shall keep light intensity at a desired light intensity level

Verification: Test and Demo.

ID: REQ-02

Description: The system shall respond to any ambient light change within max. 3 seconds

Verification: Test and Demo.

ID: REQ-03

Description: The user shall be able to configure the system using a terminal

Rational: In order to set the desired light intensity in lux

Verification: Testing and demo

System Requirements:

ID: SYS-01

Description: The user shall be able to interact with system through a terminal via serial

Rationale: To provide the user a simple interface to configure the desired light intensity.

Acceptance Criteria:

- The serial communication shall be configured as 115200-8-N-1.
- The user shall be able to send commands from the terminal to the system
- The system shall correctly receive and process the commands and provide corresponding responses through the terminal.

ID: SYS-02

Description: The system shall be able to store/restore the configuration values to/from the system non-volatile memory.

Rationale: To ensure that the configured system settings are retained across power cycles and can be restored when required.

Acceptance Criteria:

- The system shall be able to store the configuration values in the non-volatile memory when the configuration is changed.
- The system shall be able to restore the configuration values from the non-volatile memory.

- The restored configuration values shall be identical to the values that were saved.

ID: SYS-03

Description: At startup, the system shall automatically restore the stored <desired_ambient_light_intensity> from the non-volatile memory.

Rationale: To ensure that the user's configured target ambient light intensity is retained and automatically applied after a system restart or power cycle.

Acceptance Criteria:

- If a valid stored value is available, the system shall restore it as the current <desired_ambient_light_intensity>.
- The restored value shall be used by the system for light-intensity control after startup.
- If no stored value is available, or an error occurs while restoring the value, the system shall indicate the error by blinking the lamp every second.

ID: SYS-04

Description: The system shall prompt the user to enter the desired light intensity value (lux) in the terminal.

Rationale: To allow the user to configure the target light intensity that the system shall attempt to maintain in the room.

Acceptance Criteria:

- The system shall display a prompt in the terminal requesting the desired light intensity value in lux.
- The user shall be able to enter a numerical light intensity value through the terminal.

ID: SYS-05

Description: The system shall read the user desired light intensity value in lux from the terminal, validate and store it in the non-volatile memory.

Rationale: To allow the user to configure the target light intensity that the system shall attempt to maintain.

Acceptance Criteria:

- The desired light intensity value entered by the user shall be considered as valid if it is within the range of [50, 1500].

- If the entered value is valid, the system shall store it in the non-volatile memory and use it as the current <desired_ambient_light_intensity> and display a success message in the terminal.
- If the entered value is not valid or an error occurs while storing the value it in the non-volatile memory, the system shall display an error message in the terminal.

ID: SYS-06

Description: Every 50 ms, the system shall read the ambient light intensity value(lux) from the light sensor and store it

in a circular buffer and take average of samples in the buffer as the <ambient_light_intensity>.

Rationale: To provide a stable ambient light intensity value by averaging multiple samples and reducing the effect of measurement noise.

Acceptance Criteria: Capacity of the circular buffer shall be 8.

ID: SYS-07

Description: The system shall adjust <lamp_light_intensity> by one step based on the difference between <ambient_light_intensity> and

<desired_ambient_light_intensity>. If <ambient_light_intensity> is below <desired_ambient_light_intensity> - <ERROR_MARGIN>,

the system shall increase <lamp_light_intensity> by one step, else If <ambient_light_intensity> is above

<desired_ambient_light_intensity> + <ERROR_MARGIN>, the system shall decrease <lamp_light_intensity> by one step, otherwise,

the system shall not change <lamp_light_intensity>.

Rationale: To maintain the light intensity close to the desired value while preventing unnecessary lamp adjustments caused by small measurement variations.

Acceptance Criteria:

- The system shall select the lamp intensity adjustment step such that the system responds to an ambient light intensity change within max. 3 seconds.
- If the lamp is completely on and the desired light intensity cannot be achieved, the system shall blink the lamp at 100% light intensity every second.
- If the lamp is completely off and the desired light intensity cannot be achieved, the system shall blink the lamp at 25% light intensity every second.